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Sentiment and Emotion Recognition from Speech to Enhance Human-Robot Interaction

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Abstract

Enabling machines to recognize human emotion from speech can significantly improve the human-robot interaction experience. Previous works on speech emotion recognition (SER) focused on the architecture design of deep learning models, and few studies on speech features combined different features through simple concatenation. This project proposes a new speech feature fusion method based on the attention mechanism to realize information transfer between different features. Three types of attention blocks are designed to implement attention-based feature fusion. The proposed method achieves up to 1.10% improvements in emotion recognition accuracy compared to current state-of-the-art models on the benchmark emotion recognition dataset IEMOCAP.

Keywords: Speech emotion recognition, human-robot interaction, deep learning, convolutional neural network, attention mechanism, multi-feature fusion

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Contents

Abstract 4
Acknowledgments 5
1 Introduction 7
1.1 Speech Features 7
1.2 Classifiers 9
1.3 Aims and Contributions 12
2 Theories and Proposed Methods 13
2.1 Theories 13
2.1.1 Convolutional Neural Network 13
2.1.2 Multiscale Feature Extraction 13
2.1.3 Attention Mechanism 14
2.2 Proposed Methods 14
2.2.1 Feature Extraction 15
2.2.2 Multiscale Feature Extraction Model 16
2.2.3 Attention Mechanism for Cross-modal Feature Fusion 16
3 Experiments and Results 19
3.1 Experimental Settings 19
3.1.1 Datasets 19
3.1.2 Comparison Models 19
3.1.3 Evaluation Metrics 19
3.1.4 Training, Validation, and Test Sets Construction 20
3.1.5 Parameters Settings 21
3.2 Results and Discussions 21
3.2.1 Comparison with SOTA SER Models 21
3.2.2 Comparison of proposed models 25
3.2.3 Comparison of different feature combinations 26
3.2.4 Effect of proposed feature fusion methods on feature embeddings 26
3.2.5 Ablation Study 28
4 Conclusions and future work 31
References 32
Appendix A 36

1 Introduction

Emotions are an integral part of human beings. They play an essential role in daily communication by prompting better mutual understanding. By endowing machines with the ability of emotional recognition, the human-robot interaction experience can be further improved. Therefore, automatic emotion recognition has been a research focus in artificial intelligence. Numerous studies have been carried out in recognizing human emotion, from gestures, and facial expressions, to speech. Among them, speech emotion recognition (SER) focuses on identifying emotional content in speech. SER received much research attention in the last two decades and remained a challenging topic. Challenges in SER arise from two aspects: the extraction of features relevant to emotion from speech signals [1, 2], and the design of suitable classifier type and architecture [3]. The following two sections review the typical features and classifiers used in SER.

1.1 Speech Features

1.1.1 Spectral Features

Spectral features are those extracted from the frequency domain. According to [4], the vocal tract characteristics of speakers are well represented in the spectral domain, past studies [5, 6] have established that those vocal track parameters can provide cues for emotion.

Typical spectral features include Linear Prediction Coefficient (LPC), Linear Prediction Cepstral Coefficients (LPCC), Mel Frequency Cepstral Coefficients (MFCC), and Gammatone Frequency Cepstral Coefficients (GFCC). MFCC is most widely used in emotion recognition [7]. To obtain MFCC, audio is first divided into frames. Fourier transform is then performed on each frame to get the frequency spectrum. Energies on several sub-bands specified by the Mel-filter bank are calculated, and inverse Fourier transform is performed on the logarithm of those energies to obtain MFCC. Many studies suggested spectral features, especially MFCC, can yield higher recognition accuracy than prosodic and voice quality features [8, 9].

Zero crossing rate (ZCR) is another type of spectral feature. It presents the rate at which the audio waveform crosses the zero axis. Research has shown the correlation between ZCR and emotion and its effectiveness in SER [10, 11].

1.1.2 Prosodic Features

Prosodic features are long-term features representing the characteristics of large units in speech, such as syllables, words, phrases, and sentences. Prosodic features are proven to reflect distinctive properties of emotional content in speech [12].

Prosodic features can be categorized into three major categories: pitch, intensity, and intonation. Among them, pitch, also known as fundamental frequency, is found to be tightly related to emotions and is most widely used in SER [13, 14]. The fundamental frequency is formed through the vibration of the vocal cord. The changes in fundamental frequency contain much information related to emotion. For example, while anger leads to a decreasing trend, happiness usually gives rise to a rising pattern.

1.1.3 Voice Quality Features

Voice quality features are the representation of the physical properties of the vocal tract. They can reflect the variability of speech from various aspects. For example, the jitter describes the changes in frequency; the shimmer measures the variation in amplitude; Harmonics to Noise Ratio (HNR) reveals the relative noise level. The effectiveness of voice quality features has been verified in past research. In [15], combining prosodic features with voice quality features yielded 10% higher emotion recognition accuracy than prosodic features alone.

1.1.4 Teager Energy Operator-Based Features

Teager and Kaiser developed the Teager energy operator because the formation of speech is produced from a non-linear vortex-airflow interaction [16, 17]. This operator measures the energy in speech formed by that non-linear process. TEO-based features can detect the alteration of airflow during a speech, which usually results from changes in muscle tension in stressful situations. Therefore, TEO-based features are most widely used to recognize stress-related emotions. In [18], TEO-based features were used to detect clinical depression and achieved performance superior to prosodic, spectral, and voice quality features.

Most previous studies in SER are based on MFCC for its well-proven effectiveness in emotion recognition. However, since other features also contain rich emotional information, studying the combined features is valuable. While some research was carried out on a combination of various speech features, I suggest that there are limitations to them. For example, in [19], five

different spectral features were used for SER, including MFCC, Mel-scaled spectrogram, Chromogram, Spectral contrast feature, and Tontz representation. These features were extracted from each frame and concatenated to form a 193x1 feature. Deep CNN was then applied to learn high-level representation from this combined feature. However, while this study considered multiple features, all those features belonged to the same class, which meant they were correlated. Besides, the way of feature combination failed to consider the differences in features, such as their scale and physical meaning.

In [20], the eGeMAPS feature set [21] was adopted, which contained a wide variety of audio features from spectral feature MFCC, prosodic feature fundamental frequency to voice quality features, including shimmer and jitter. Features extracted from 65 successive frames were stacked to form an 88x65 feature map. CNN was then applied to obtain high-level feature representation. While audio features other than spectral features were considered, the way of feature combination was poorly designed. Along the feature dimension, different elements have different physical meanings. For example, while MFCCs represent the energy in different bandwidths, HNR describes the ratio between the human voice and environmental noise. Convolutions on those features did not produce meaningful results. Besides, the significantly different numerical scales of distinctive features make it harder for the shared convolution kernel to learn.

This research aims to overcome the limitation of previous studies. Three ways of feature fusion are proposed to better exploit diverse types of features. To my knowledge, no previous work in SER has applied such methods.

1.2 Classifiers

Classifiers used for SER can be divided into two groups: traditional machine learning (ML) classifiers and deep learning (DL) classifiers.

1.2.1 Traditional Machine Learning Classifiers

Traditional ML algorithms include K-nearest neighbourhoods, support vector machine (SVM), naïve Bayesian (NB), artificial neural network (ANN), decision tree (DT), Hidden Markov Model (HMM), and ensemble learning. Early studies on SER using traditional ML achieved

several satisfactory results. [22, 23] leveraged SVM for SER on the Berlin Emo DB dataset (Emo-DB), obtaining 70% and 81.3% accuracy, respectively. [24] experimented on GMM, MLP, HMM, and a newly proposed hierarchical model, which yielded 63.49%, 66.83%, 68.57%, and 71.75% recognition rates, respectively.

However, bounded by simple model structures, traditional ML classifiers have limited expressive power and always fail to learn complicated patterns. For example, SVM performs well only when data from distinct groups are linearly separatable. Although kernel-SVM can alleviate this problem, choosing a good kernel is a challenging task. In SER, traditional ML classifiers perform poorly on more challenging datasets like IEMOCAP. [25] utilized NB and SVM on IEMOCAP, achieving merely 41.87% and 42.01% accuracy, while [26] employed Decision Tree, which yielded a 58.46% recognition rate.

1.2.2 Deep Learning Classifiers

Deep learning models get their names from their deep structures. They usually consist of multiple layers, which endow them with strong non-linear mapping ability, thus the capability to recognize complex patterns. Typical deep learning models include deep neural networks (DNN), deep belief networks (DBN), convolutional networks (CNN), recurrent neural networks (RNN), and attention-based Transformers. Most studies in SER were focused on CNN, RNN, and Transformer, because of their mechanisms which enable temporal information fusion.

CNNs consist of some convolutional layers. Convolution operations are performed in convolution layers to fuse regional information. For example, in image processing, convolutional operation on a patch of pixels can produce a feature representation of that whole patch so that it can be analyzed integrally. [27] leveraged Fourier-transform-based filter bank speech spectrograms as the input features, experimenting on various CNN architectures on IEMOCAP. The optimal model yielded 64.78% accuracy, achieving more than 10% absolute improvement compared to the SVM baseline. In [28], a layout of CNN for SER was proposed, which obtained 81.75% accuracy on IEMOCAP.

RNNs utilize ‘memory’ to memorize information, thus achieving temporal information fusion. They comprise memory cells that take input features at each time step and get corresponding feature representations. Gates control the amount of information to be preserved or discarded. [29] implemented a bidirectional extended short-term memory network, a type of RNN, for

SER, achieving 70.34% accuracy. [30] also employed LSTM-based RNN and reported comparable results (71.04%) on IEMOCAP.

Transformer realizes information flow through the attention mechanism. The attention mechanism allows the model to access information at various positions [31]. In [32], an SER model was proposed based on the transformer-based speech representation framework wav2vec 2.0 [33], and 78.15% accuracy was obtained on IEMOCAP.

CNN, RNN, and Transformer-based models are the most popular deep learning architectures in SER, and they have helped realize great advances in recognition accuracy. These three types of models have their own set of strengths and weaknesses.

CNN utilizes a shared convolutional kernel on each convolution layer, reducing the number of parameters, and making the model light weighted. Besides, the convolutional operation enables information fusion in local areas. However, limited by the size of convolution kernels, CNNs are insensitive to the global feature and the relationship between elements that are widely separated in location.

For RNNs, re-utilizing one memory cell for all time steps also saves memory. They are suitable for handling sequential data, especially sequence generation tasks like language translation and speech recognition. However, since only one memory unit is used, they tend to forget earlier inputs and thus also perform poorly in capturing long-term features. Moreover, because data need to be processed sequentially, RNNs have a low degree of parallelism, making training slow and computationally expensive. Finally, the vanishing gradient problem [34] makes long-term dependencies hard to learn, making model training difficult.

Transformer leverages the attention mechanism, allowing any two feature elements to attend to each other. This implements more flexible feature fusion and, thus, a stronger ability to recognize complex relationships and representing complicated patterns. However, implementing the attention mechanism makes the transformer-based models large and contain more parameters than CNN and RNN. This makes them more computationally expensive and memory-intensive to train and deploy.

1.3 Aims and Contributions

The purpose of this research project is to achieve improvement over current SOTA emotion recognition models by combining multiple speech features with MFCC. Spectral, prosodic, and voice quality features include MFCC, zero crossing rate, fundamental frequency, shimmer, and jitter. MFCC is used as the major discriminator, while other features are used as supplementary. As for the classifier, due to limited computation resources, large-scale transformer-based models are not considered. And for the choice between CNN and RNN, several experiments (Appendix A) showed that in this classification task where sequential output is not required, CNNs can fulfill the task well, whereas RNNs always fail to converge. Therefore, the CNN-based SER model is the focus of my research project. The essential backbone inherits a CNN-based model, which achieved SOTA performance in SER on the IEMOCAP dataset [35]. Based on their model framework, attention blocks with three different attention mechanisms are proposed to implement information fusion, among other features. Experiments were carried out to verify the effectiveness of my proposed attention blocks and feature fusion scheme.

Overall, the main contributions of this project are:

- A novel method of utilizing multiple speech features for emotion recognition is proposed. To my knowledge, no previous work in SER has applied such a feature combination scheme.
- Three attention blocks are designed to implement the proposed multi-feature fusion.
- Experiments demonstrate that my proposed method and models improve emotion recognition on the IEMOCAP dataset compared to current SOTA methods.

The rest of this report is organized as follows. In section 2, related theories applied in this project are first reviewed, followed by the description of the proposed methods. Section 3 presents the experimental setting and a detailed analysis of the experiment results. Finally, the conclusion and further work are discussed in Section 4.

2 Theories and Proposed Methods

2.1 Theories

2.1.1 Convolutional Neural Network

Convolutional Neural Network (CNN), a typical model backbone in deep learning, has been widely studied and particularly used in computer vision. Following the trend in computer vision, much research was carried out on the usage of CNN in other fields, such as natural language processing. In image processing, convolutional kernels enable CNN to capture spatial features, whereas, in sequential data processing like audio processing, it means the ability to obtain and synthesize long term-feature.

2.1.2 Multiscale Feature Extraction

The size of the convolutional kernel in CNN determines the coverage area of the output feature on the input feature map. By arranging multiscale convolutional kernels in parallel, the model can capture features at different scales. In speech emotion recognition, the input feature is typically obtained by stacking MFCCs of successive frames, constituting a 2D map, where two dimensions represent time and frequency, respectively. Multiscale convolution allows features within different timespan and frequency ranges to be captured separately, as shown in Figure 1.

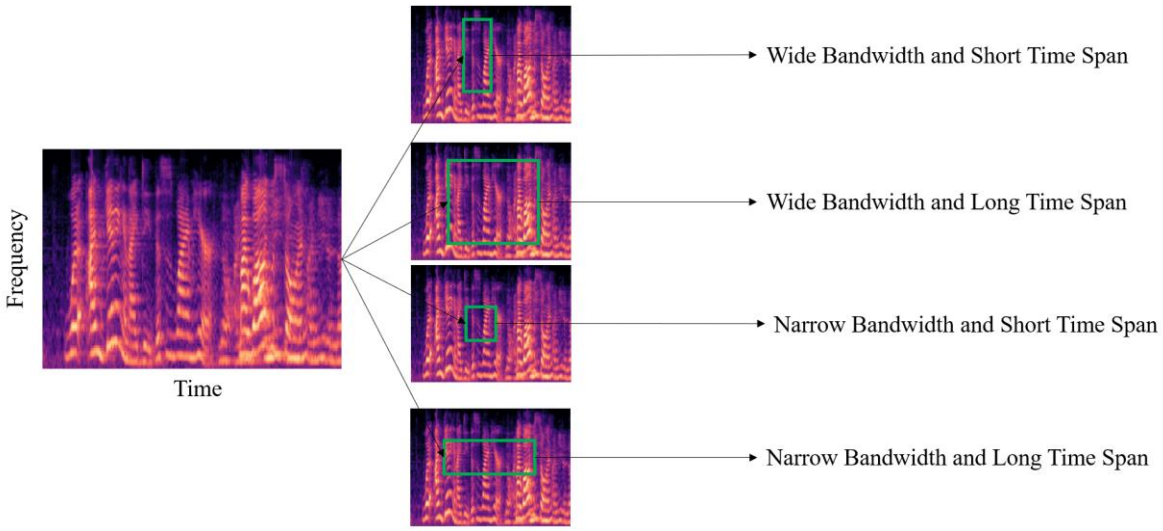


Figure 1. Illustration of multiscale feature extraction

2.1.3 Attention Mechanism

Since being introduced in 2017 [31], the attention mechanism has received great attention and has been extensively used in various deep-learning models. It addresses the problem faced by traditional deep learning models. While the size of the convolution kernel limits CNN and emphasizes too much on the local area, the forgetting mechanism in RNN makes them prone to lose long-term information. In Comparison, the attention mechanism allows any two feature units at any time step or location to influence each other, breaking the limitation in both space and time. For example, in image processing, a pixel on an object tends to pay more attention to pixels constituting that object and ignore other patches. At the same time, in language translation, words will attend to other words that are semantically related regardless of their position. Overall, the attention mechanism can realize the fusion of global features.

Apart from global feature fusion, the attention mechanism can also be used to achieve information fusion across multiple different feature representations. Figure 2 illustrates an example of attention-based feature fusion. After applying cross-attention, both feature maps integrate each other's information.

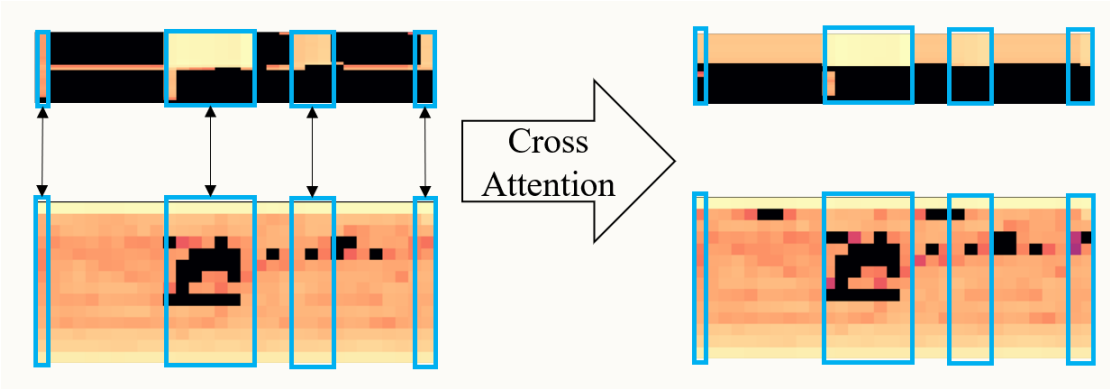


Figure 2. Example of using attention mechanism for cross-feature information fusion

2.2 Proposed Methods

This section presents a CNN-based cross-modality attention framework for speech emotion recognition. Unlike most off-the-shelf CNN-based speech recognition models that solely utilize spectral features, particularly MFCC, as input features, the proposed framework adopts MFCC as the major differentiating feature, together with zero crossing rate, shimmer, jitter, and fundamental frequency as supplementary features. For simplicity, zero crossing rate, shimmer, jitter, and fundamental frequency are abbreviated as *auxiliary features*, while MFCC is named as the *main feature*. Two parallel model pipelines with similar structures are used

separately for processing the auxiliary and main features. Three attention-based cross-modality information-exchanging schemes are proposed to fuse the auxiliary feature and the main features at each level. The overall model structure is shown in Figure 3. The backbones of the model, including multiscale blocks and global aware blocks, inherit from the model proposed in [35], while the arrangements of a parallel pipeline for auxiliary features and attention block for cross-modality feature fusion are my proposals. Details about model architecture are provided in the following section.

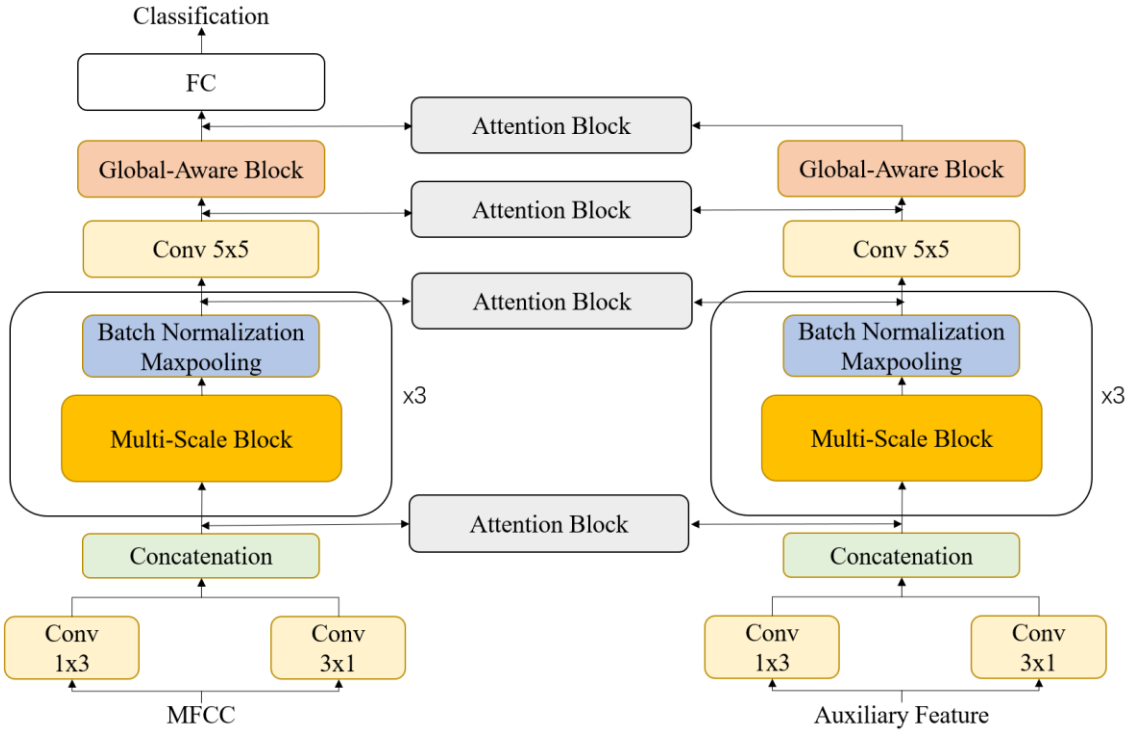


Figure 3. Proposed multi-feature speech emotion recognition model. Two CNN networks with the same architectures process two feature streams in parallel. Attention blocks are used to exchange information between two feature streams.

2.2.1 Feature Extraction

As stated above, spectral, voice quality, and prosodic features are used as model input. Audios are sampled at 16000Hz and divided into several 128ms frames at a step of 32ms. Speech features are then extracted from each frame. For the main spectral feature, MFCC is chosen for its well-proven effectiveness in SER. As for the auxiliary feature, the fundamental frequency reflects prosodic characteristics, while shimmer and jitter are used for voice quality representation. Zero crossing rate, another spectral feature effective in SER, is also used as an auxiliary. Python Library Librosa [36] and OpenSmile [37] are employed to extract these

features. Features of T successive frames are stacked to form $\mathrm{D_Y} \times \mathrm{T}$ auxiliary feature map and main feature map $D_X \times T$. In my project, 26-MFCC is used, and features from 63 successive frames are combined, thus $D_X = 26$, T=63. The auxiliary feature is a subset of the fundamental frequency, shimmer, jitter, and zero crossing rate, so $\mathrm{D_Y} \in \{1,2,3,4\}$, indicating the number of auxiliary features.

2.2.2 Multiscale Feature Extraction Model

The first level of the model consists of two multiscale shallow feature extraction blocks, each of which has two convolution kernels. For the main feature $X \in R^{1\times D_X \times T}$, there is one 1x3 convolution kernel to obtain temporal information and another 3x1 kernel for frequency domain information fusion. Both convolution operations lead to $\mathrm{C} \times \mathrm{D_X} \times \mathrm{T}$ outputs, which are then concatenated to form a $\mathrm{C} \times 2\mathrm{D_X} \times \mathrm{T}$ feature. As for auxiliary feature $Y \in R^{1\times \mathrm{D_Y} \times \mathrm{T}}$, pre-processing is done before convolution for consistency with operation on the main feature. It is worth noting that while the second dimension $\mathrm{D_X}$ represents different frequency ranges, $\mathrm{D_Y}$ is the number of features. Therefore, I reshape Y into $D_Y \times 1 \times T$ and perform 1x3 and 1x1 convolution separately. After convolution, two $\mathrm{C} \times 1 \times T$ tensors are obtained. These tensors are replicated along the second dimension for $\mathrm{D_X}$ times and then concatenated to form a $\mathrm{C} \times 2\mathrm{D_X} \times \mathrm{T}$ feature.

In subsequent levels, a similar block structure is adopted. Each multiscale block consists of parallel 3x3 and 5x5 convolution kernels, which implement multiscale feature extraction. Max pooling and batch normalization are applied following each block.

2.2.3 Attention Mechanism for Cross-modal Feature Fusion

The multiscale feature extraction model introduced above serves as the backbone to process the main feature and auxiliary feature in two separate models. The main feature model is primarily responsible for SER, while the auxiliary feature provides supplementary information to assist this process. The attention mechanism is implemented to achieve information communication between these two streams of features. Three different methods of using attention for this purpose are proposed in the following sections.

2.2.3.1 Cross-Modality Attention Weighing

The first scheme **C**ross-**M**odality **A**ttention **W**eighing (CMAW) uses auxiliary features for determining significant regions to attend to on the main feature map. Specifically, assuming at the n-th level, for an auxiliary feature map with a size of $\mathrm{C_n} \times \mathrm{D_n} \times T_n$, a 1x1 convolution is applied and $C_n \times \mathrm{D_n} \times T_n$ weight tensor is obtained. This weight tensor is then multiplied by the main feature map to obtain the weighted feature, as shown in Figure 4. Intuitively, this mechanism can work because those auxiliary features like zero crossing rate and fundamental frequency can indicate whether the human voice is present in that frame. With this attention mechanism, the most significant and informative features can be emphasized.

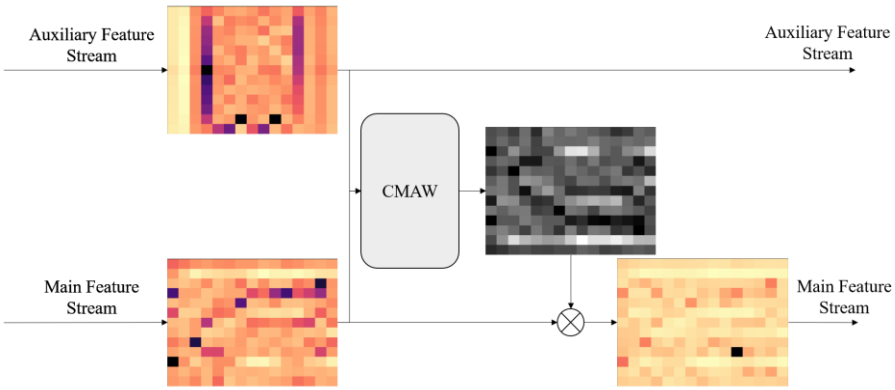


Figure 4. Illustration of Cross-Modality Attention Weighing (CMAW) block

2.2.3.2 Cross-Modality Unidirectional Attention

In the first method, the auxiliary feature is only used to generate a weight map to filter out irrelevant features and emphasize crucial ones. In the second scheme, Cross-Modality Unidirectional Attention (CMUA), information in the auxiliary feature is fused into the main feature through the attention mechanism. Three 1x1 convolutions project $\mathrm{C_n} \times \mathrm{D_n} \times T_n$ auxiliary feature into $\mathrm{H_n} \times \mathrm{D_n} \times T_n$ *key* and $\mathrm{C_n} \times \mathrm{D_n} \times T_n$ *value*, and main feature into $\mathrm{H_n} \times \mathrm{D_n} \times T_n$ *query*. By computing the dot product of the *query* and *key* in the first dimension, an attention score map of size $1 \times \mathrm{D_n} \times T_n$ is obtained. Multiplying the attention score map with *value*, $\mathrm{C_n} \times \mathrm{D_n} \times T_n$ supplementary feature is obtained, which is then added to the main

feature map to complete feature fusion. Figure 5 illustrates the mechanism and effect of the CMUA block.

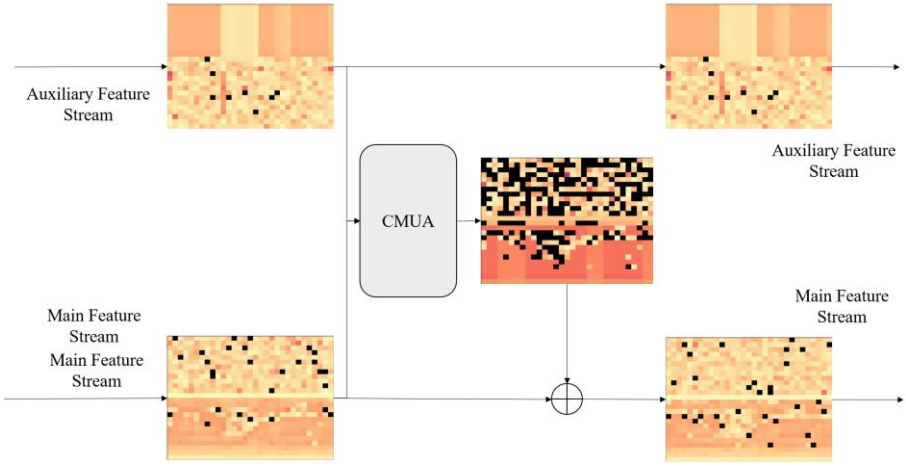


Figure 5. Illustration of Cross-Modality Unidirectional Attention (CMUA) block

2.2.3.3 Cross-Modality Bidirectional Attention

The final way of feature fusion **C**ross-**M**odality **B**idirectional **A**ttention (CMBA) is through bidirectional attention. The implementation is like the second scheme. The only difference is that: while the auxiliary feature is fused into the main feature, information in the main feature is also transmitted to the auxiliary feature, as shown in Figure 6. In other words, it realizes communication between auxiliary and main feature streams.

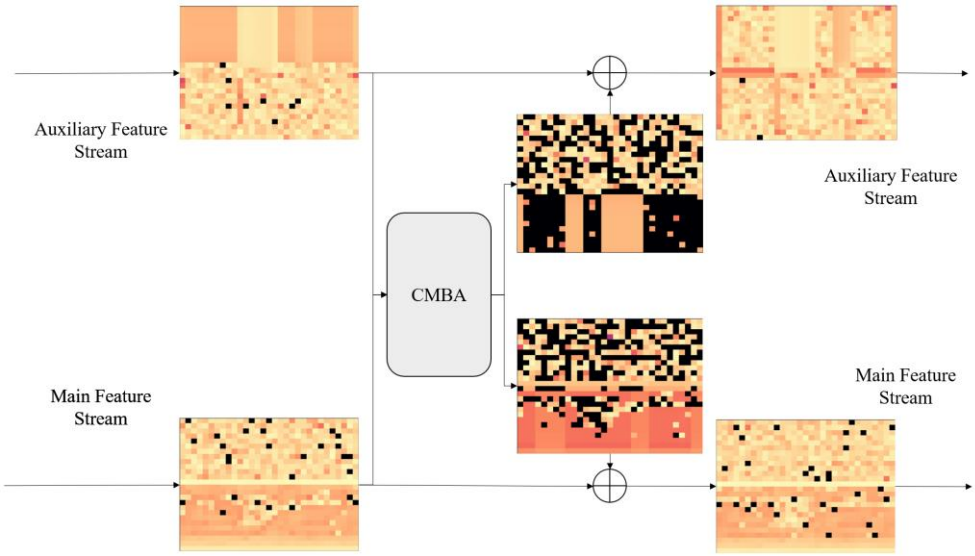


Figure 6. Illustration of Cross-Modality Bidirectional Attention (CMBA) block

3 Experiments and Results

3.1 Experimental Settings

3.1.1 Datasets

This section evaluates the proposed auxiliary feature enhancement model on the IEMOCAP dataset, which contains 12 hours of situational dialogues performed by 10 actors with nine different emotions. In this experiment, to be comparable with previous research, five basic types of emotions, neutral, angry, sad, excited, and happy, are selected. Happy and excited are labeled happy for their similarity and consistency with previous studies. The processed dataset contains 4084 audio samples. Among them, there are 1320 neutral samples, 967 happy samples, 948 sad samples, and 849 angry samples.

3.1.2 Comparison Models

To prove the effectiveness of proposed attention blocks, Area Attention CNN (AACNN) [38] and Global-Aware Multiscale (GLAM) neural network [35], which achieved SOTA performance on SER is used as a benchmark model for comparison. AACNN and GLAM utilize MFCC as the input feature. In contrast, proposed models CMAW, CMUA, and CMBA use the combination of MFCC and auxiliary speech features, including ZCR, fundamental frequency, shimmer, and jitter. To demonstrate the contribution of these auxiliary features to emotion recognition and find out the best combination, 15 different feature sets were considered, covering all the combinations of these features. These 15 feature sets are used separately on CMAW, CMUA, and CMBA, leading to 45 model-feature combinations. Overall, a total of 47 cases were tested and compared.

3.1.3 Evaluation Metrics

Six different metrics were used for model performance evaluation, including micro accuracy (weighted accuracy WA), micro recall (weighted recall WR), micro F1 score, macro accuracy (unweighted accuracy UA), macro recall (unweighted recall UR), and macro F1 score. Accuracy measures the percentage of correct predictions among all predictions, while recall describes the portion of correctly recognized samples in all samples. The F1 score is the harmonic mean of accuracy and recall. Micro accuracy (recall, f1 score) is obtained from the weighted average of accuracies (recalls, f1 scores) of all classes, where the weight is the percentage of samples of each class. Macro accuracy (recall, f1 score) is simply the unweighted

average of all accuracies (recalls, f1 scores). The mathematical expressions of above metrics are shown as follows.

For class i where i∈{1,2,3,4,5,6}, we assume that the percentage of class i is w_i, the number of samples that belong to i and are correctly predicted is TP_i, the number of samples that belong to i but are falsely predicted as other classes is FN_i, the number of samples that do not belong to i but are falsely predicted as i is FN_i, the number of samples that do not belong to i and predicted as other classes is TN_i. Then the weighted accuracy WA, unweighted accuracy UA, weighted recall WA, unweighted recall UR, micro f1, and macro f1 can be expressed as:

$$WA = \frac{\sum_{i=1}^{6} TP_i}{\sum_{i=1}^{6} TP_i + FP_i} \tag{1}$$

$$UA = \frac{\sum_{i=1}^{6} \frac{TP_i}{TP_i + FP_i}}{6} \tag{2}$$

$$WR = \frac{\sum_{i=1}^{6} TP_i}{\sum_{i=1}^{6} TP_i + FN_i} \tag{3}$$

$$UR = \frac{\sum_{i=1}^{6} \frac{TP_i}{TP_i + FN_i}}{6} \tag{4}$$

$$W - F1 = \sum_{i=1}^{6} w_i F1_i \tag{5}$$

$$U - F1 = \frac{\sum_{i=1}^{6} F1_i}{6} \tag{6}$$

3.1.4 Training, Validation, and Test Sets Construction

MFCC, ZCR, fundamental frequency, shimmer, and jitter are extracted frame-by-frame to form training and test samples. Each signal is first sampled at 16000Hz, 128ms frames were then obtained at 32ms. This means for any two consecutive frames, there are 96ms overlaps. On each frame, above mentioned speech features are extracted using the Python library Librosa [36] and OpenSmile [37]. 26-MFCC is used in this case, which yielded 30-dimensional features. 63 successive feature vectors form one sample $\in R^{30\times63}$. Samples are taken at intervals of 12 frames. While the complete feature map $\in R^{30\times63}$ is fed as input into models, models obtain feature selectively on the map. For benchmark models, AACNN and GLAM, only last 26 dimensions, namely the MFCC map $\in R^{26\times63}$ is used as input feature. For proposed models,

except for the MFCC map, auxiliary features specified in the feature set is extracted from the remaining 4 dimensions and input into the parallel block.

Training, validation, and test sets were split at the ratio of 8:1:1. To obtain convincing results, splits were done with 5 different random seeds, and experiments were carried out on all cases. The average performance was taken to evaluate different models and feature combinations.

3.1.5 Parameters Settings

Parameter settings for model training and evaluation are as follows. The batch size is set to 32. Cross-entropy loss is used as the objective function to train and evaluate the models. Adam optimizer is employed with a weight decay rate of 10^{-6}. The technique of learning rate decay is adopted. Initialized to 10^{-4}, the learning rate decreases exponentially with a factor of 0.95 until reaching 10^{-6}. Each model is trained for 50 epochs.

3.2 Results and Discussions

3.2.1 Comparison with SOTA SER Models

Experiments were carried out on the IEMOCAP dataset with the settings mentioned above. The results are shown in Tables 1, 2, and 3, corresponding to the three proposed feature fusion methods. Results indicate that using proposed feature fusion models on certain feature combinations can improve emotion recognition accuracy. Specifically, CMAW performs best when used with fundamental frequency and zero-crossing rate, achieving around 1.0% and 5.0% improvement on all metrics compared to GLAM and AACNN, as shown in Table 1. Table 2 shows that CMUA obtains its optimal performance when using the combination of fundamental frequency and shimmer as auxiliary features, reaching 0.5% and 4.5% higher precision and recall rates than GLAM and AACNN. From Table 3, we can observe that CMBA performs best with shimmer and jitter. It outperforms GLAM by 0.8% in unweighted accuracy and recall and 0.2% in weighted accuracy and recall.

Method	WA	WR	UA	UR	W-F1	U-F1
AACNN	76.41±0.32	75.59±0.28	78.48±1.06	72.93±0.37	74.92±0.09	75.50±0.21
GLAM	80.56±1.23	80.12±1.28	81.77±0.63	77.86±0.58	79.36±0.58	79.95±1.19
CMAW F	81.11±1.30	**80.74±1.20**	**82.93±1.57**	79.24±0.70	**80.59±1.10**	80.54±1.24
CMAW J	80.99±0.76	80.74±0.85	82.26±0.52	78.48±1.08	79.89±0.90	80.48±0.86
CMAW S	80.60±1.62	80.50±1.56	80.55±1.69	**79.83±0.93**	80.01±1.20	80.38±1.51
CMAW Z	**81.23±0.94**	**80.84±0.85**	82.11±1.28	78.89±1.17	80.02±0.80	**80.67±0.86**
CMAW FJ	80.52±1.98	80.35±1.98	81.42±2.03	79.55±1.53	80.22±1.84	80.17±2.05
CMAW FS	80.61±1.21	80.50±1.20	81.22±1.19	79.02±1.71	79.91±1.40	80.44±1.22
CMAW JS	80.37±1.42	79.95±1.49	81.92±1.46	78.41±0.64	79.71±0.99	79.85±1.41
CMAW FZ	**81.64±1.70**	**81.45±1.77**	82.41±1.84	**80.14±1.48**	**81.10±1.64**	**81.40±1.71**
CMAW JZ	81.00±1.34	80.58±1.20	**82.46±1.65**	78.44±0.78	79.96±1.05	80.46±1.18
CMAW SZ	**81.31±0.84**	**80.74±1.06**	82.44±0.98	**79.75±2.24**	**80.59±1.24**	**80.67±1.11**
CMAW FJS	80.96±1.67	80.50±1.56	81.59±2.38	78.84±0.93	79.68±1.31	80.34±1.50
CMAW FJZ	80.31±0.43	80.35±0.50	80.84±0.11	79.45±0.68	79.99±0.38	80.20±0.48
CMAW FSZ	80.22±1.10	79.80±1.27	80.92±1.45	78.27±1.15	79.14±1.17	79.64±1.24
CMAW JSZ	80.60±0.84	80.27±0.99	81.70±0.44	78.65±1.92	79.77±1.32	80.15±1.02
CMAW FJSZ	80.90±1.39	80.42±1.42	**82.78±1.52**	78.46±1.42	80.20±1.40	80.36±1.42

Table 1. Performance comparison of proposed Cross-Modality Attention Weighing (CMAW) Block with AACNN and GLAM. AACNN and GLAM use MFCC as the feature, while CMAW employs the combination of MFCC with auxiliary speech features. 15 different auxiliary feature combinations are tested separately. F, J, S, and Z stand for fundamental frequency, jitter, shimmer, and zero-crossing rate respectively. For metrics, WA, WR, UA, UR, W-F1, and U-F1 stand for weighted accuracy, weighted recall, unweighted accuracy, unweighted recall, weighted f1 score, and unweighted f1 score, respectively. The top 3 performances for each metric are highlighted.

Method	WA	WR	UA	UR	W-F1	U-F1
AACNN	76.41±0.32	75.59±0.28	78.48±1.06	72.93±0.37	74.92±0.09	75.50±0.21
GLAM	**80.56±1.23**	**80.12±1.28**	81.77±0.63	77.86±0.58	**79.36±0.58**	**79.95±1.19**
CMUA F	79.96±1.12	79.36±1.06	81.87±0.99	77.73±0.84	79.28±0.93	79.21±1.08
CMUA J	79.75±0.21	79.36±0.35	81.36±0.22	77.12±0.09	78.70±0.03	79.19±0.30
CMUA S	79.35±0.22	78.66±0.64	80.90±1.03	76.90±0.84	78.35±0.21	78.57±0.68
CMUA Z	79.72±1.12	79.13±1.06	81.75±0.95	76.38±0.68	78.25±0.74	78.93±1.08
CMUA FJ	79.88±1.05	79.13±1.06	82.25±0.95	77.03±1.02	79.10±0.99	79.11±1.08
CMUA FS	**81.08±0.34**	**80.42±0.42**	**83.37±0.06**	**77.94±0.39**	**79.99±0.31**	**80.32±0.44**
CMUA JS	79.75±0.73	79.01±0.42	82.21±1.01	77.07±0.11	78.96±0.31	78.84±0.45
CMUA FZ	**80.57±0.46**	**79.72±0.14**	**83.04±0.78**	**78.00±0.66**	**79.78±0.16**	**79.53±0.11**
CMUA JZ	79.91±1.41	79.36±1.20	82.00±1.63	77.19±0.59	79.04±0.92	79.25±1.17
CMUA SZ	80.28±1.39	79.25±1.13	83.15±1.55	76.64±0.71	79.01±0.94	79.10±1.09
CMUA FJS	79.57±1.18	79.01±1.27	81.20±1.18	77.76±0.55	79.01±0.85	79.00±1.22
CMUA FJZ	79.46±0.72	78.54±0.71	81.97±0.88	76.82±0.23	78.75±0.48	78.58±0.64
CMUA FSZ	80.14±1.06	79.72±1.13	**82.32±0.89**	78.16±1.19	79.72±1.11	**79.53±1.16**
CMUA JSZ	80.19±0.61	79.48±0.85	82.29±0.32	77.38±1.01	79.28±0.72	79.39±0.79
CMUA FJSZ	79.51±1.30	79.36±1.34	80.32±1.46	**78.24±0.99**	79.04±1.23	79.26±1.32

Table 2. Performance comparison of proposed Cross-Modality Unidirectional Attention (CMUA) Block with AACNN and GLAM. AACNN and GLAM use MFCC as the feature, while CMAW employs the combination of MFCC with auxiliary speech features. 15 different auxiliary feature combinations are tested separately. F, J, S, and Z stand for fundamental frequency, jitter, shimmer, and zero-crossing rate respectively. For metrics, WA, WR, UA, UR, W-F1, and U-F1 stand for weighted accuracy, weighted recall, unweighted accuracy, unweighted recall, weighted f1 score, and unweighted f1 score, respectively. The top 3 performances for each metric are highlighted.

Method	WA	WR	UA	UR	W-F1	U-F1
AACNN	76.41±0.32	75.59±0.28	78.48±1.06	72.93±0.37	74.92±0.09	75.50±0.21
GLAM	80.56±1.23	80.12±1.28	81.77±0.63	77.86±0.58	79.36±0.58	79.95±1.19
CMBA F	80.56±0.78	**80.25±0.85**	81.43±0.81	**78.66±1.16**	**79.61±0.56**	**80.07±0.70**
CMBA J	80.21±0.29	79.72±0.19	81.80±0.12	77.91±0.62	79.49±0.44	79.62±0.11
CMBA S	79.39±1.10	78.69±0.66	81.15±2.26	75.31±0.68	77.34±1.20	78.43±0.77
CMBA Z	79.90±0.70	79.32±0.66	81.90±0.87	77.23±0.73	79.08±0.73	79.21±0.61
CMBA FJ	**80.63±0.54**	79.56±0.38	**83.09±0.82**	76.66±0.34	79.03±0.41	79.44±0.41
CMBA FS	79.27±0.77	78.77±0.85	80.91±0.67	76.17±1.13	77.93±1.11	78.64±0.89
CMBA JS	79.66±0.38	79.01±0.19	81.83±1.07	77.24±1.36	78.96±0.64	78.87±0.27
CMBA FZ	80.28±0.62	79.87±0.57	82.04±0.79	77.73±0.70	79.39±0.48	79.74±0.51
CMBA JZ	80.27±0.61	79.56±0.57	**82.56±0.77**	76.77±0.90	78.96±0.87	79.37±0.58
CMBA SZ	**80.83±1.11**	**80.35±0.85**	82.49±1.60	**78.32±0.25**	**79.94±0.75**	**80.26±0.90**
CMBA FJS	80.57±1.39	80.03±1.70	81.95±1.53	**78.13±2.38**	79.63±1.91	80.01±1.76
CMBA FJZ	**80.78±0.48**	**80.35±0.47**	**82.56±0.68**	78.08±0.47	**79.76±0.56**	**80.15±0.54**
CMBA FSZ	80.03±1.15	79.48±0.94	81.95±1.70	77.00±1.04	78.90±1.29	79.35±0.94
CMBA JSZ	80.22±0.42	79.87±0.47	81.43±0.97	77.65±1.05	79.13±0.57	79.71±0.50
CMBA FJSZ	80.23±0.38	79.95±0.28	81.52±0.93	78.27±1.40	79.46±0.55	79.78±0.33

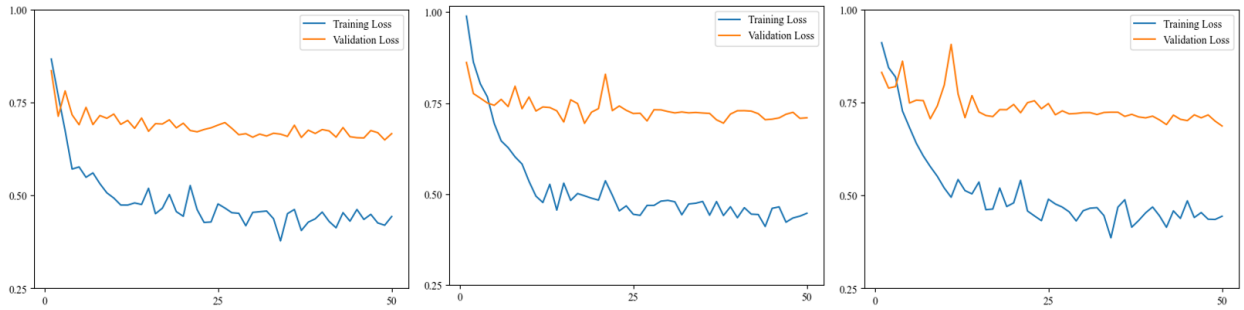
Table 3. Performance comparison of proposed Cross-Modality Bidirectional Attention (CMBA) blocks with AACNN and GLAM. AACNN and GLAM use MFCC as the feature, while CMAW employs the combination of MFCC with auxiliary speech features. 15 different auxiliary feature combinations are tested separately. F, J, S, and Z stand for fundamental frequency, jitter, shimmer, and zero-crossing rate respectively. For metrics, WA, WR, UA, UR, W-F1, and U-F1 stand for weighted accuracy, weighted recall, unweighted accuracy, unweighted recall, weighted f1 score, and unweighted f1 score, respectively. The top 3 performances for each metric are highlighted.

3.2.2 Comparison of proposed models

The top 2 performances achieved by each of the three proposed models are presented in Table 4. Overall, CMAW has the best performance among the three proposed models. This is probably because CMAW has the simplest architecture and thus is less inclined to overfit. Figure 7 compares the training and validation loss curve of three proposed models CMAW, CMUA, and CMBA, from which we can see that CMUA and CMBA suffer a severer overfitting problem than CMAW.

Method	WA	WR	UA	UR	W-F1	U-F1
AACNN	76.41±0.32	75.59±0.28	78.48±1.06	72.93±0.37	74.92±0.09	75.50±0.21
GLAM	80.56±1.23	80.12±1.28	81.77±0.63	77.86±0.58	79.36±0.58	79.95±1.19
CMAW FZ	**81.64±1.70**	**81.45±1.77**	**82.41±1.84**	**80.14±1.48**	**81.10±1.64**	**81.40±1.71**
CMAW SZ	**81.31±0.84**	**80.74±1.06**	**82.44±0.98**	**79.75±2.24**	**80.59±1.24**	**80.67±1.11**
CMUA FS	81.08±0.34	80.42±0.42	83.37±0.06	77.94±0.39	79.99±0.31	80.32±0.44
CMUA FZ	80.57±0.46	79.72±0.14	83.04±0.78	78.00±0.66	79.78±0.16	79.53±0.11
CMBA SZ	80.83±1.11	80.35±0.85	82.49±1.60	78.32±0.25	79.94±0.75	80.26±0.90
CMBA FJZ	80.78±0.48	80.35±0.47	82.56±0.68	78.08±0.47	79.76±0.56	80.15±0.54

Table 4. Comparison between proposed methods CMAW, CMUA, CWBA, and benchmark methods AACNN and GLAM. The top two performances are highlighted.



a) CMAW FZ b) CMUA FS c) CMBA SZ

Figure 7. Comparison of training and validation loss curves among three proposed methods.

3.2.3 Comparison of different feature combinations

From the features perspective, their influence on emotion recognition depends on the feature fusion model being used. In general, fundamental frequency, shimmer, zero-crossing rate, and their combinations are the most effective auxiliary features in improving emotion recognition accuracy. Figure 8 presents the confusion matrixes of GLAM and CMAW with different feature combinations. Comparing b, c, d, and e with a, we can observe the effect of each feature. Fundamental frequency contributes significantly to the recognition of sadness and slightly to happiness; jitter is useful in recognizing neutral and happiness; shimmer and zero-crossing rate can also improve the recognition accuracy of neutral and happiness but degrade the identification rate of sadness and anger. When those four features are used jointly as shown in Figure 8. f), the model performs better at all emotions except for anger. The overall recognition accuracy is improved by around 1%.

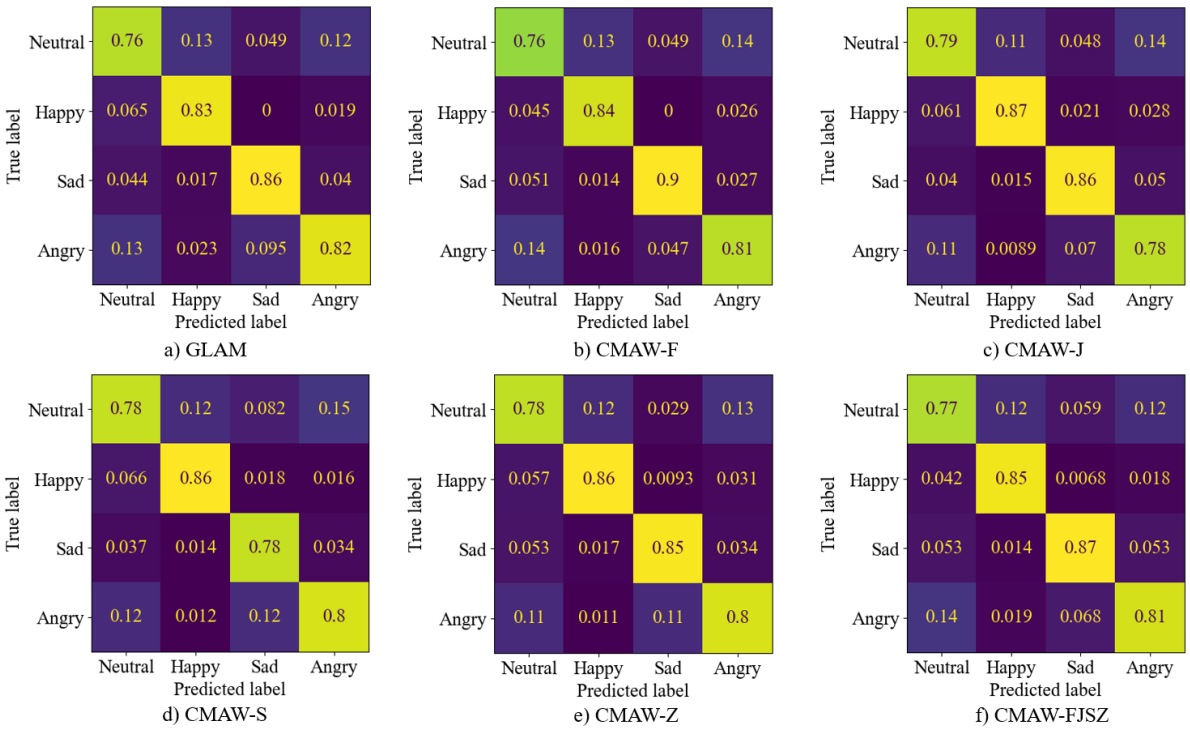


Figure 8. Comparison of GLAM and CMAW with different feature combinations. F, J, S, and Z stand for fundamental frequency, jitter, shimmer, and zero-crossing rate, respectively.

3.2.4 Effect of proposed feature fusion methods on feature embeddings

To better visualize the effect of the proposed methods, the feature embeddings obtained from proposed models and benchmark models GLAM are compared. T-distributed stochastic neighbor embedding is performed to map high-dimensional embedding feature vectors to 2D

space. Comparing a) with b), c), and d) in Figure 9, we can see that information fusion between auxiliary features and the main feature stream contributes to the feature embedding learning: the clustering of emotions is more significant and there are few overlaps between different emotions. This can contribute to separating different emotions and, thus better recognition accuracy.

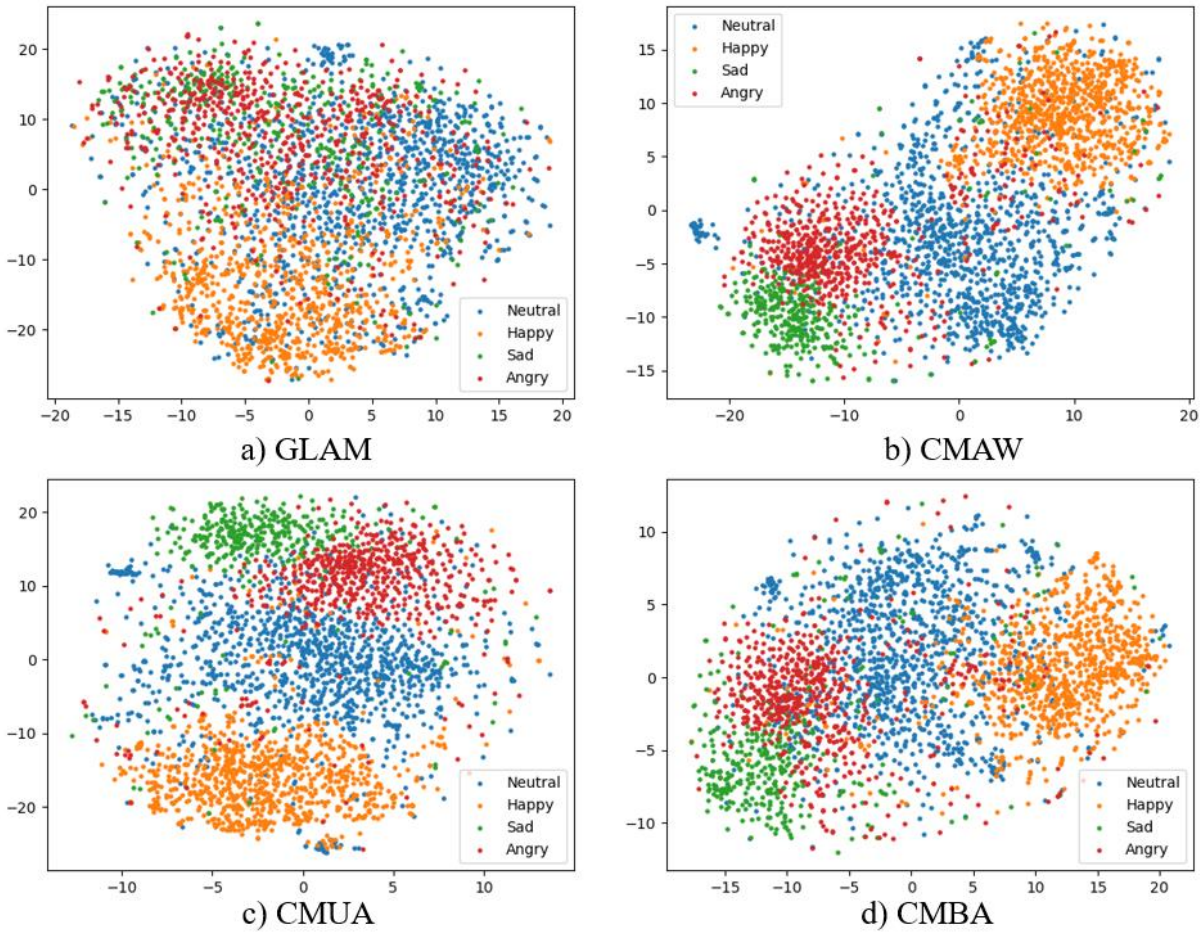


Figure 9. T-SNE of feature embeddings obtained from a) GLAM, b) CMAW, c) CMUA, and d) CMBA.

3.2.5 Ablation Study

The ablation experiment was conducted to verify the significance of my proposed attention-based feature-fusion blocks. In this experiment, instead of using the proposed attention-based methods, feature fusion was implemented through simple concatenation: $D_Y \times T$ auxiliary feature map was concatenated with $D_X \times T$ MFCC map to form $(D_X + D_Y) \times T$ input feature map, where D_X=26 and $D_Y \in \{1,2,3,4\}$. GLAM, the CNN framework of proposed CMAW, CMUA, and CMBA, was utilized as the model to process the concatenated speech features and obtain emotion labels. Experiment results are shown in Tables 5 and 6. From Table 5, we can observe that adding additional features without using proposed feature fusion methods even degrades the performance. Table 6 demonstrates the usefulness of the proposed attention-based feature-fusion blocks.

Method & Features	WA	WR	UA	UR	W-F1	U-F1
AACNN	76.41±0.32	75.59±0.28	78.48±1.06	72.93±0.37	74.92±0.09	75.50±0.21
GLAM	80.56±1.23	80.12±1.28	81.77±0.63	77.86±0.58	79.36±0.58	79.95±1.19
GLAM F	**77.06±1.07**	**76.73±1.13**	**77.93±1.08**	**74.79±1.40**	**75.85±1.23**	**76.57±1.08**
GLAM J	75.92±0.75	75.94±0.85	76.29±0.46	74.20±0.75	75.03±0.63	75.80±0.81
GLAM FJ	76.19±1.01	75.86±1.06	77.26±0.88	73.54±1.04	75.03±0.99	75.73±1.10
GLAM S	75.62±0.25	75.55±0.28	75.70±0.82	74.53±0.65	74.94±0.75	75.47±0.28
GLAM FS	**76.65±1.06**	**76.42±0.85**	77.47±1.38	**74.47±0.78**	75.61±1.00	**76.26±0.92**
GLAM JS	76.44±0.11	75.86±0.14	77.32±0.59	74.11±0.50	75.14±0.37	75.66±0.22
GLAM FJS	75.88±1.16	75.55±1.06	76.29±1.61	73.25±0.88	74.28±1.11	75.29±1.05
GLAM Z	75.13±0.63	75.08±0.57	75.62±0.85	73.27±0.62	74.28±0.72	74.96±0.60
GLAM FZ	76.58±0.42	76.34±0.50	**77.97±0.36**	74.27±1.03	**75.68±0.78**	76.17±0.50
GLAM JZ	76.58±0.47	76.18±0.35	77.05±0.57	73.80±0.76	75.02±0.43	76.01±0.31
GLAM FJZ	75.26±0.98	75.16±0.85	75.87±1.59	72.73±0.50	73.86±0.95	74.91±0.86
GLAM SZ	**75.65±0.37**	75.55±0.42	76.31±0.54	73.59±0.99	74.62±0.68	75.34±0.45
GLAM FSZ	74.69±0.61	74.61±0.57	75.23±0.69	73.48±1.27	73.93±0.96	74.30±0.68
GLAM JSZ	75.73±1.76	75.55±1.77	76.89±1.58	73.20±1.42	74.55±1.44	75.35±1.74
GLAM FJSZ	76.43±1.82	76.18±1.77	77.05±1.82	74.44±2.43	75.36±2.12	76.02±1.80

Table 5. Performance of GLAM model on directly concatenated speech features. Auxiliary features are directly concatenated with MFCC to form input features. 15 different auxiliary feature combinations are tested separately. F, J, S, and Z stand for fundamental frequency, jitter, shimmer, and zero-crossing rate respectively. For metrics, WA, WR, UA, UR, W-F1, and U-F1 stand for weighted accuracy, weighted recall, unweighted accuracy, unweighted recall, weighted f1 score, and unweighted f1 score, respectively. The top 3 performances for each metric are highlighted.

Method	WA	WR	UA	UR	W-F1	U-F1
GLAM F	77.06±1.07	76.73±1.13	77.93±1.08	74.79±1.40	75.85±1.23	76.57±1.08
GLAM FS	76.65±1.06	76.42±0.85	77.47±1.38	74.47±0.78	75.61±1.00	76.26±0.92
CMAW FZ	**81.64±1.70**	**81.45±1.77**	**82.41±1.84**	**80.14±1.48**	**81.10±1.64**	**81.40±1.71**
CMAW SZ	**81.31±0.84**	**80.74±1.06**	**82.44±0.98**	**79.75±2.24**	**80.59±1.24**	**80.67±1.11**
CMUA FS	81.08±0.34	80.42±0.42	83.37±0.06	77.94±0.39	79.99±0.31	80.32±0.44
CMUA FZ	80.57±0.46	79.72±0.14	83.04±0.78	78.00±0.66	79.78±0.16	79.53±0.11
CMBA SZ	80.83±1.11	80.35±0.85	82.49±1.60	78.32±0.25	79.94±0.75	80.26±0.90
CMBA FJZ	80.78±0.48	80.35±0.47	82.56±0.68	78.08±0.47	79.76±0.56	80.15±0.54

Table 6. Comparison of proposed methods CMAW, CMUA, CWBA, and GLAM. The top 2 performances are selected for each model.

4 Conclusions and future work

In this project, a combination of multiple speech features for SER is proposed. Unlike previous works, which either consider solely MFCC or directly concatenate various features, an attention-based feature fusion method is used to achieve information fusion among multiple feature streams. Three different model designs are further proposed to implement such an information fusion scheme. The experimental results show that the proposed method could achieve a 1.0% improvement over the current SOTA SER model on the IEMOCAP dataset. For future studies, new methods and model architectures for feature fusion could be explored to employ multiple speech features jointly for emotion recognition.

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Appendix A

In the middle stage of my project, I experimented with a wide variety of self-build deep learning models, in particular CNN and RNN. For CNN, I first tested traditional CNNs, including AlexNet, ResNet, and Inception Net on SAVEE, and they yielded 51.9%, 43.6%, and 46.7% weighted accuracy, respectively. The confusion matrix for AlexNet is shown below.

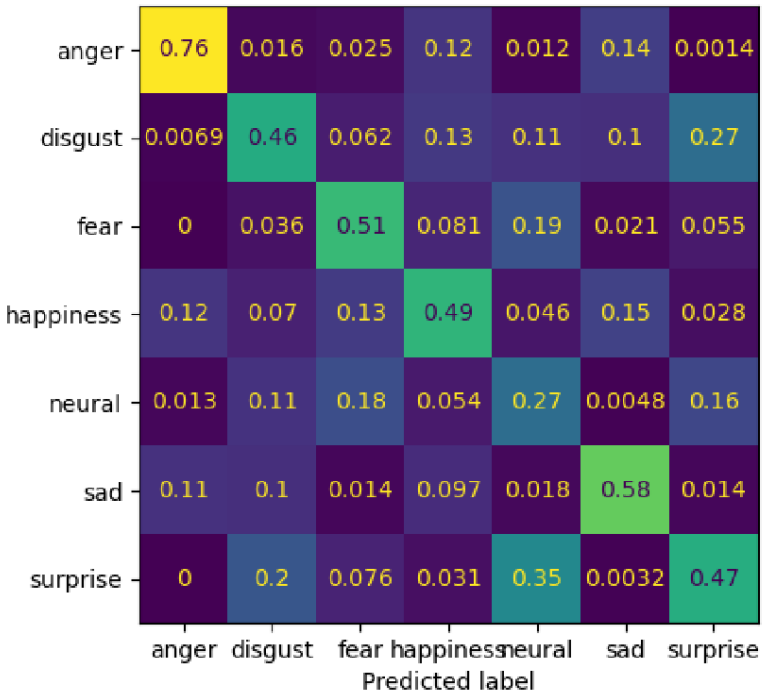

Figure 10. The result obtained by AlexNet on SAVEE

Above CNNs only accept fixed-length inputs, which means utterances must be analyzed on the frame level. In another scheme, multiple-scale convolution kernels were utilized to obtain features that cover different time spans. Adaptive pooling was applied to select the most salient features and to enable the model to handle variable length inputs so that MFCC maps extracted from complete utterances can be directly used as inputs. On SAVEE, the best performance was 68.8% accuracy, achieved when kernel lengths in the first layer are 1, 2, 4, 8, 16, 32, and 64.

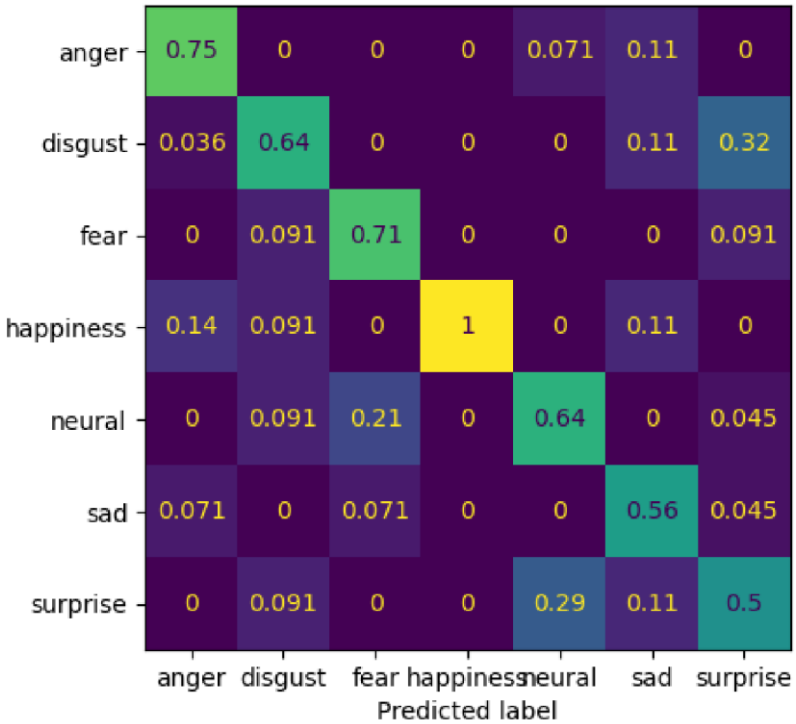

Figure 11. The result obtained by my multiscale CNN on SAVEE

As for RNN, I experimented with LSTM using MFCC sequence as the input feature. Various parameter settings were tested, and no model obtain satisfactory results. All models would reach a state of predicting only on a single label.